High-Fidelity Quantum Logic Gates Using Trapped-Ion Hyperfine Qubits

Joshua Tensuan and Bryant Har

High-Fidelity Quantum Logic Gates Using Trapped-Ion Hyperfine Qubits, Ballance et al. (2016)

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Context and Background

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Quantum Computing

DIVINCENZO CRITERIA

REQUIREMENTS FOR THE PHYSICAL IMPLEMENTATION OF QUANTUM COMPUTATION	
D1: Scalable qubits	Scalable physical system of well-defined, characterized qubits
D2: Initialization	Prepare a simple, fiducial input state
D3: Measurement	Measure the qubit state
D4: Universal gate set	Perform a universal set of gate operations with high fidelity
D5: Coherence	Robustly represent quantum information (long coherence times)
REQUIREMENTS FOR ROUTING QUANTUM INFORMATION	
D6: Interconversion	Ability to interconvert stationary and flying qubits
D7: Communication	Ability to transmit flying qubits faithfully between two locations

Image Source: https://so.ilitchbusiness.wayne.edu/knowledge/what-is-the-divincenzo-criteria

Paradigms of Quantum Computation

Platform	Qubit	Pros	Cons
Superconducting	Circuit energy states	Fast gates, scalable, mature	Short coherences, cryogenic cooling

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Trapped-ion Quantum Computers

Error Rates still too high

- Benhelm et al. (2008): Fidelity of 99.3%
- Seek lower error for fault-tolerance

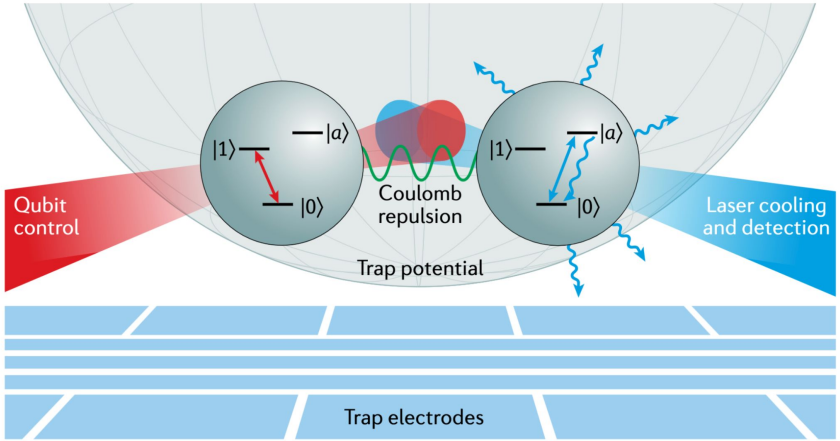

Image: Chiaverini et al, Nat Rev Mater (2021).

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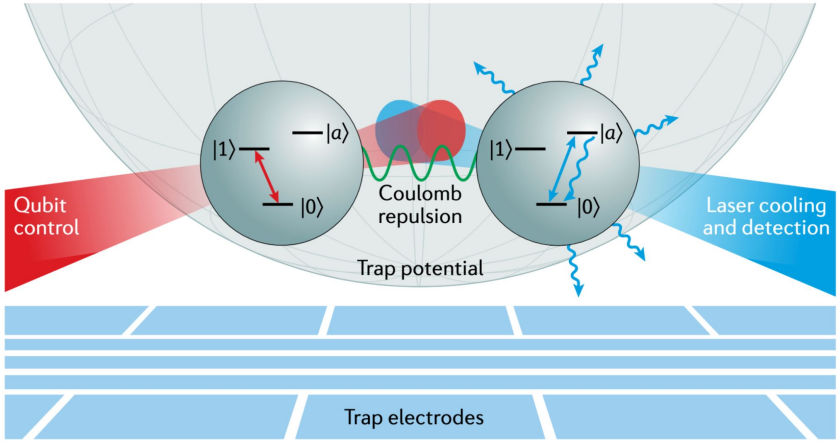
Gate Speeds

- Fast compared to coherence time
- Tradeoff with fidelity



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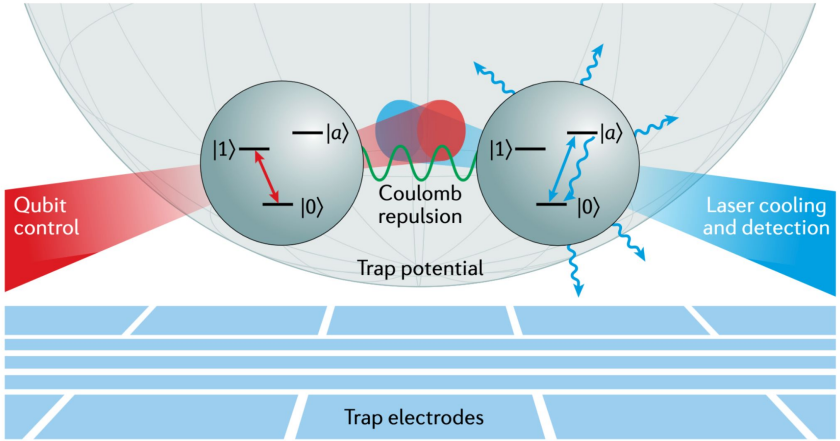
Theory well-established

- Leibfried et al. (2003), Home et al. (2006)



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Research Goal

- Experimentally demonstrate geometric phase 2-qubit gate
- Analyze and optimize trade-off between gate speed and fidelity
- Characterize sources of error

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Experimental Design

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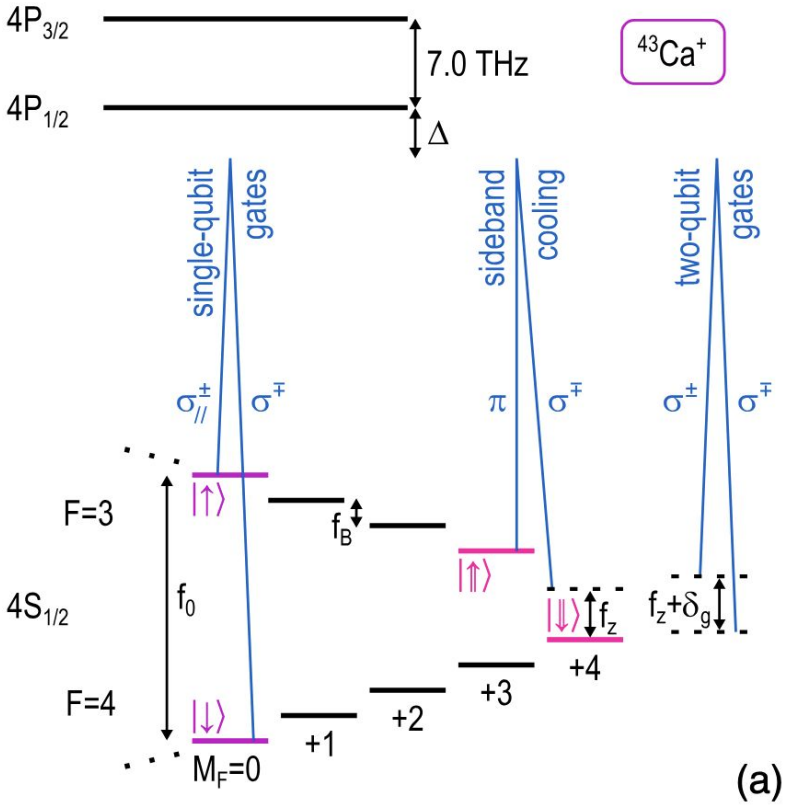
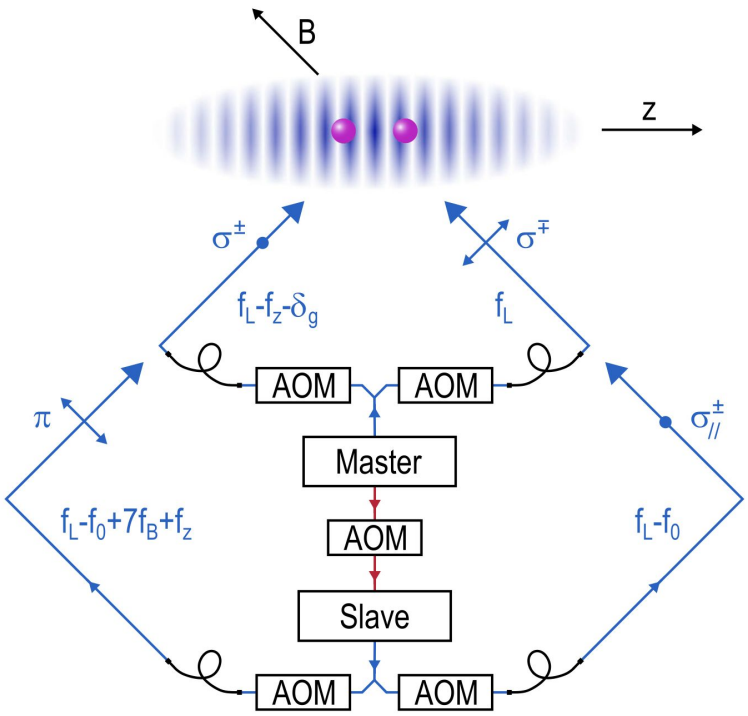
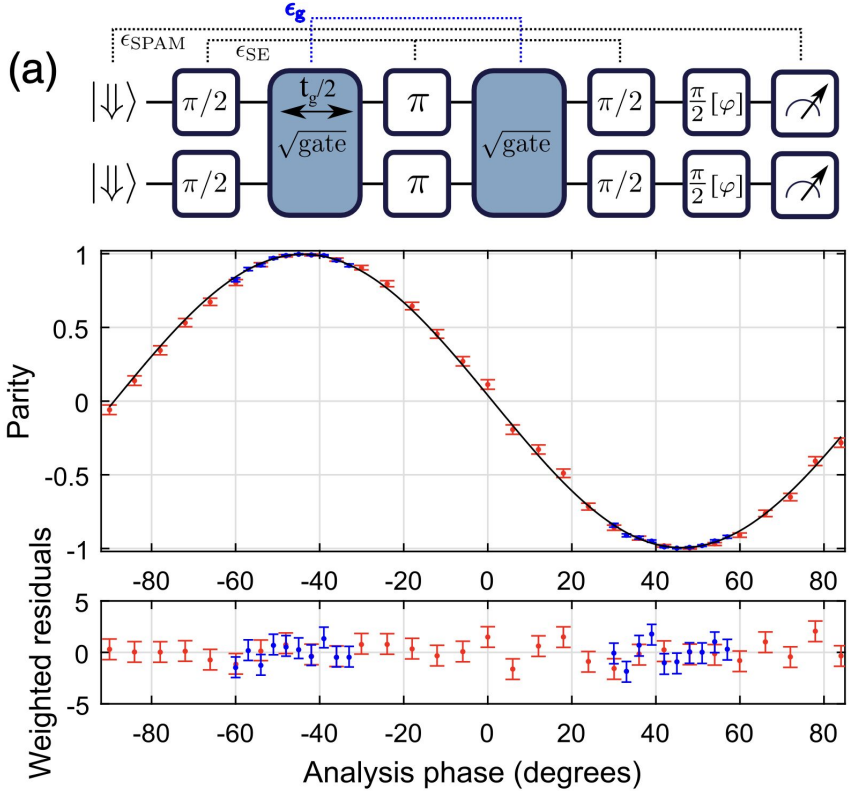
Calcium Ion Level Structure

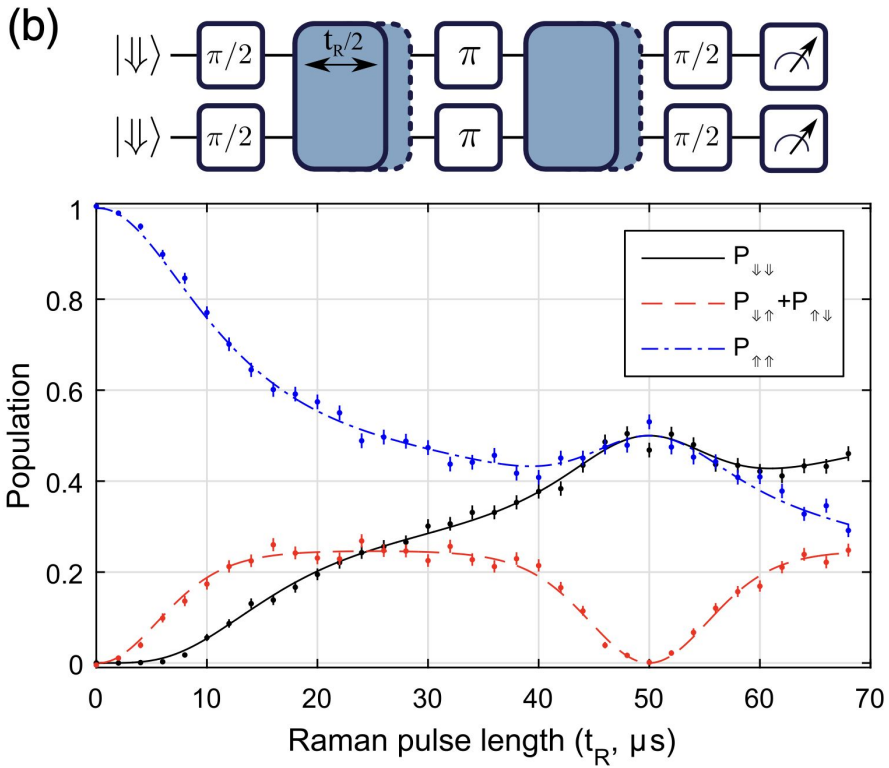




Two-Qubit Quantum State Tomography



Time-Dependent Dynamics of Gate Operation



Analysis

Infidelity Measurements and Error Modeling

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Theoretical Error Modeling

Two-qubit gate error source (at $t_g = 100\mu s, \Delta = -3.0$ THz)	Calculated error ($\times 10^{-3}$)
Raman + Rayleigh photon scattering	0.4
Motional heating and dephasing	0.2
Spin dephasing	0.2
Raman beam intensity drift ($<0.5\%$)	<0.06
Motional temperature ($\bar{n} < 0.05$)	<0.04
Off-resonant effects	<0.01
Total	0.9

Dominant Sources of Error Modeled

Each error calculation follows from supplemental material and Ballance's PhD thesis (2014).

For example, error from Raman scattering (gamma as rate) is approximated for small errors as:

$$\epsilon_{\text{Raman}} = \frac{3}{2}\Gamma_{\text{Raman}} t$$

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Components of Theoretical Model

Source of Error	Description
Raman Scattering	Scattering of light with a different frequency (light-matter interactions)
Rayleigh Scattering	Scattering of light with same frequency as incident
Motional Heating, Dephasing	Unwanted kinetic energy in trapped ions/Quantum coherence decaying over time
Spin Dephasing	Loss of coherence in spin states (decay of quantum information)
Raman Beam Intensity Drift	Fluctuations in the intensity of Raman laser beams over time.

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Dominant at low gate times (faster gate speeds)

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Dominant at high gate times (slower gate speeds)

Components of Theoretical Model

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Raman Beam Intensity Drift	Fluctuations in the intensity of Raman laser beams over time.

Intuition: Low gate times require higher intensity beams, so effects magnify.

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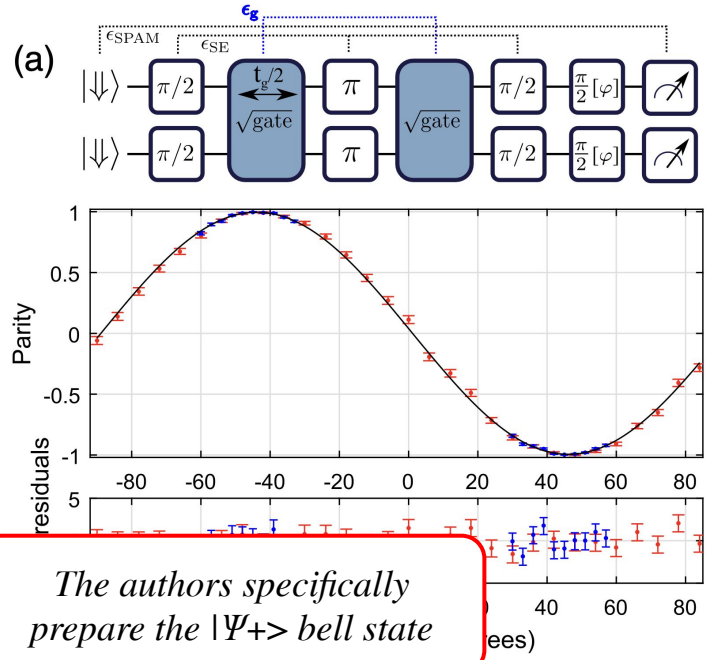
Intuition: High gate time magnify time-dependent effects (e.g. dephasing).

Fidelity via Quantum Tomography Procedure

As a quick refresher, we retrieve the density matrix ρ using quantum state tomography by sweeping phi in the circuit to the right.



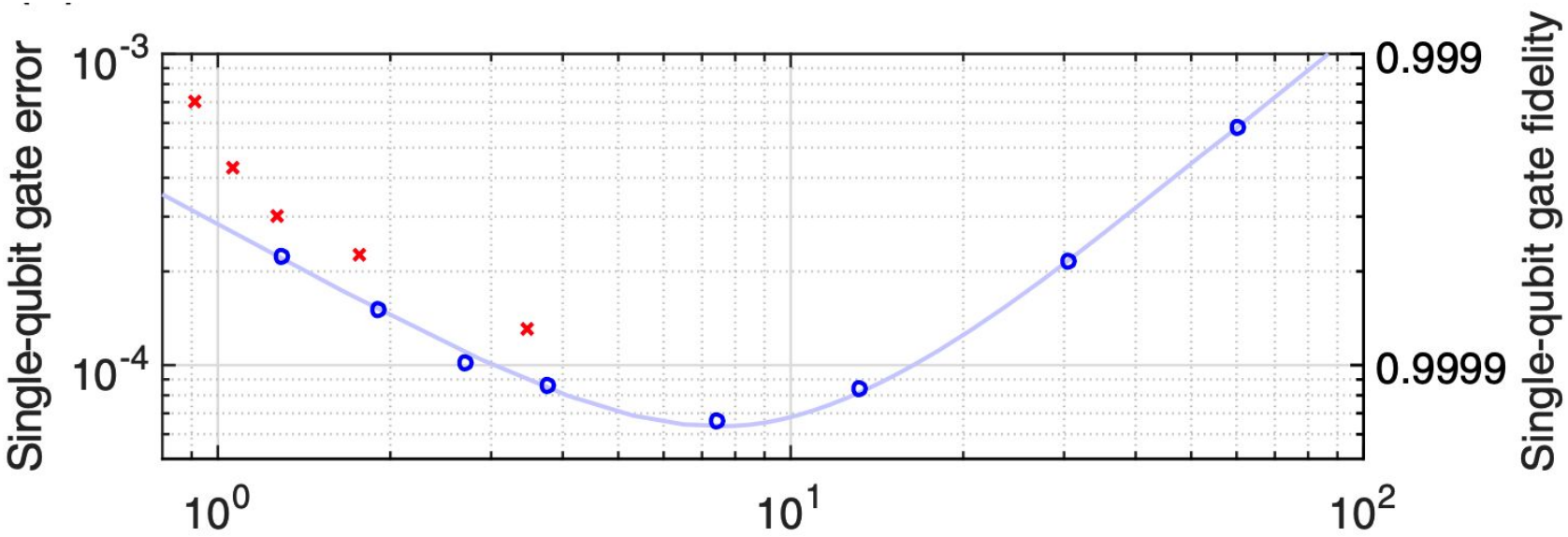
With the ρ density matrices, fidelity can be computed with the bell state |Ψ+> that we prepare

$$F = \langle \psi_+ | \rho | \psi_+ \rangle$$

$$|\psi_+\rangle = (|\Downarrow\Downarrow\rangle + |\Uparrow\Uparrow\rangle)/\sqrt{2}$$

The authors specifically prepare the |Ψ+> bell state

Fidelity (Single-qubit Gate)



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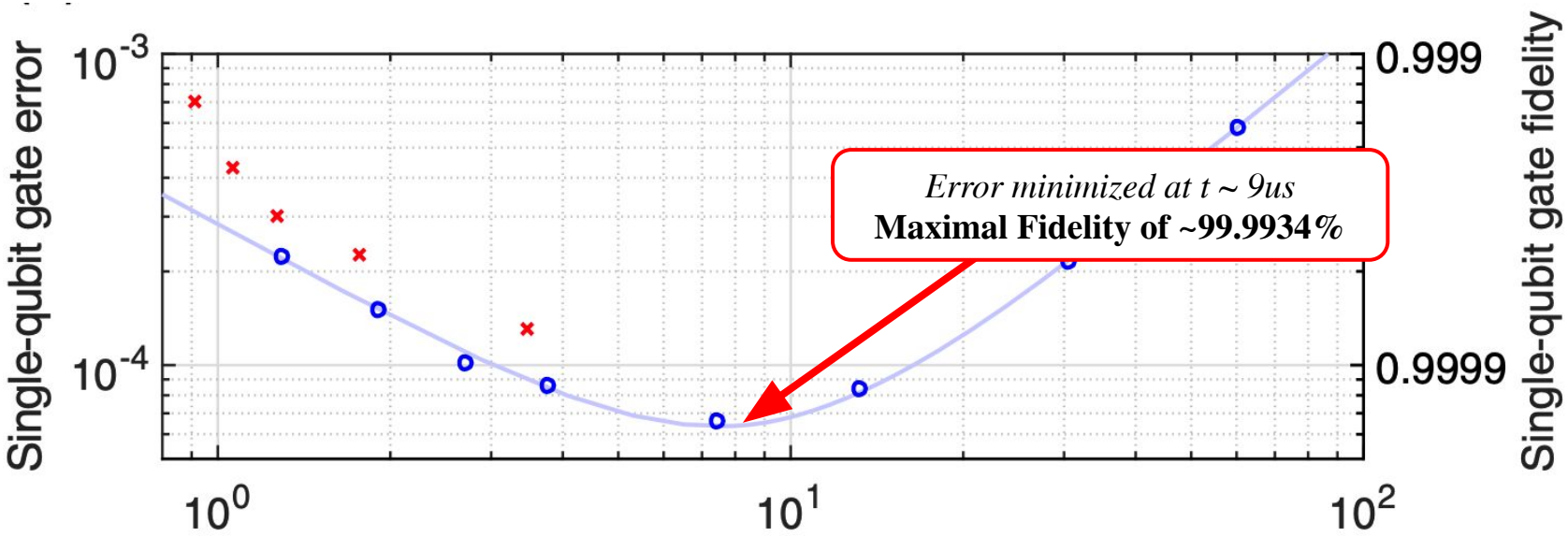
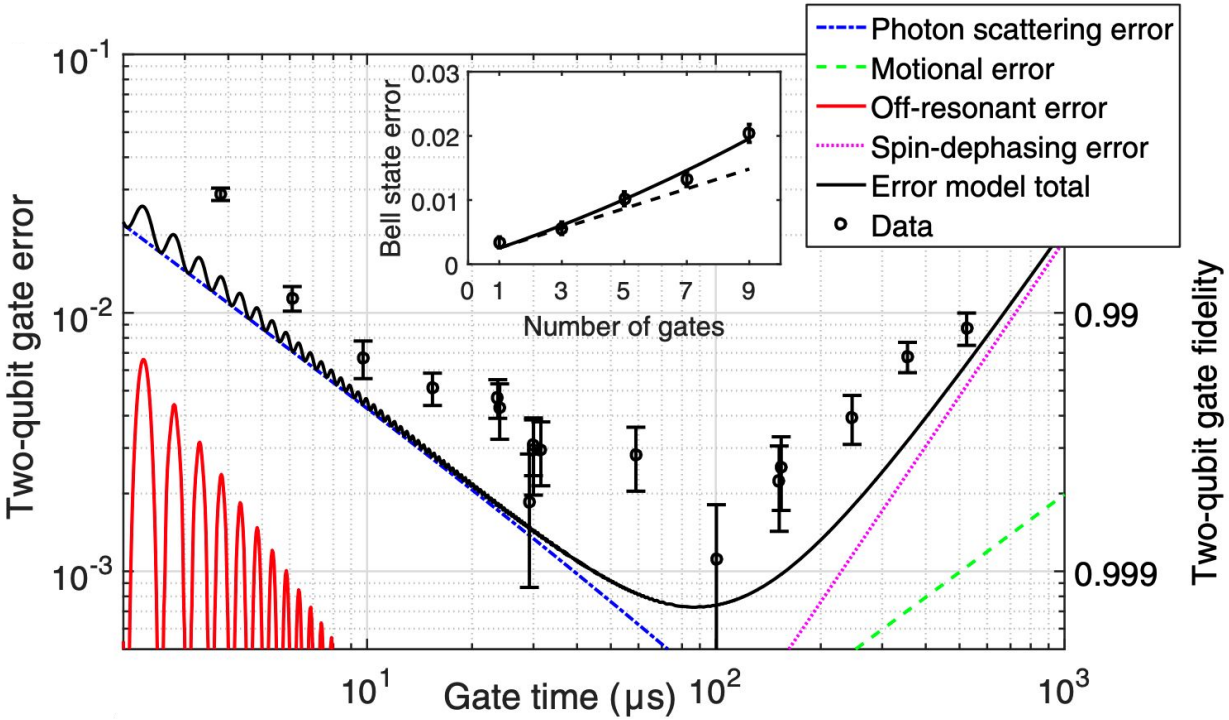


Fidelity (Two-qubit Gate)



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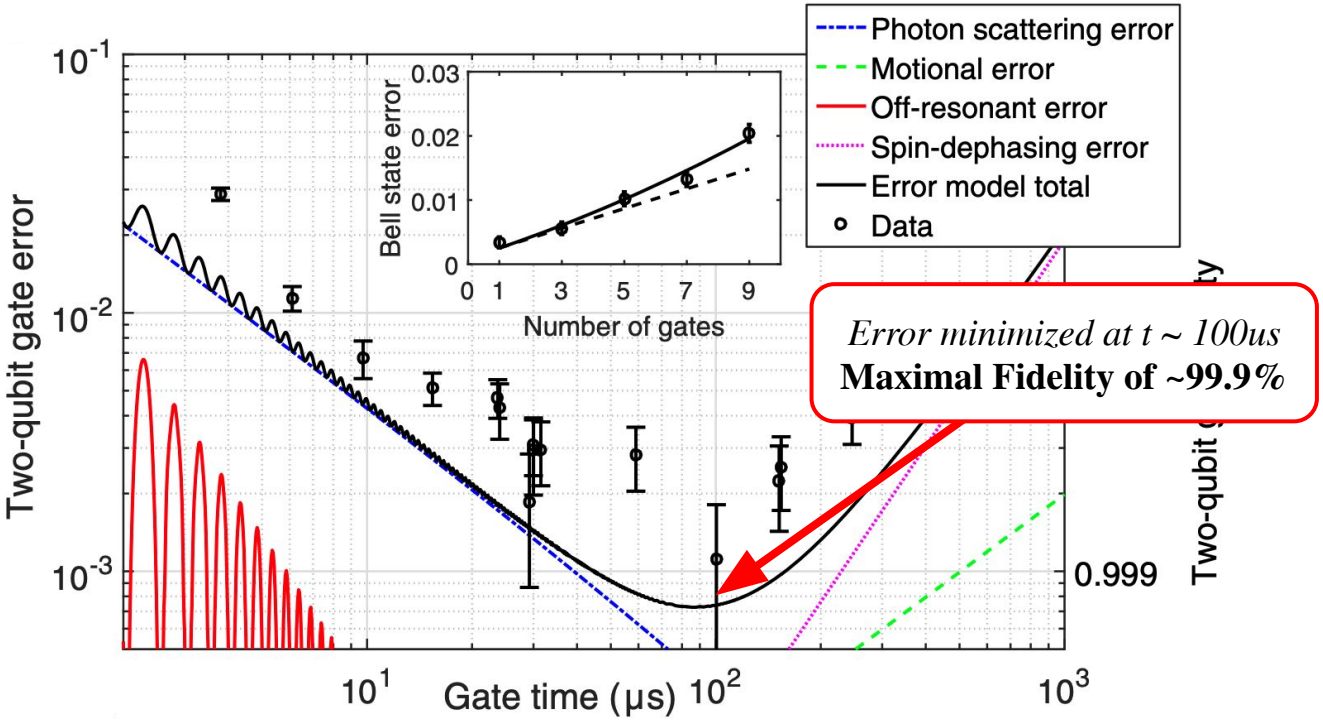

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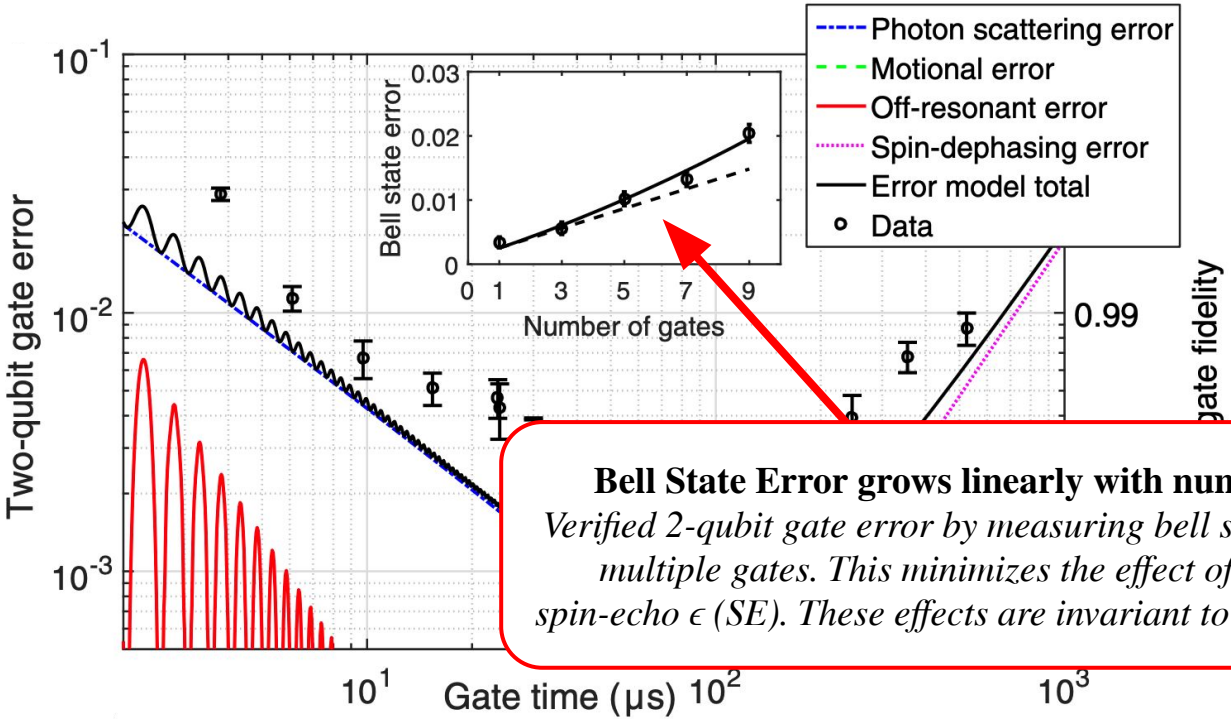

Bell State Error grows linearly with number of gates

Verified 2-qubit gate error by measuring bell state error across multiple gates. This minimizes the effect of SPAM ϵ and spin-echo ϵ (SE). These effects are invariant to number of gates.

Conclusion

Future Experiments and Impact

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Implementation of Gates

Two proposed methods to integrate one and two-qubit laser gates in a scalable manner

Quantum CCD vs **Network Model**

Implementation of the Gates

Quantum CCD

Inspired by classical charge-coupled device design

Ions are shuttled around a large microfabricated array of interconnected traps

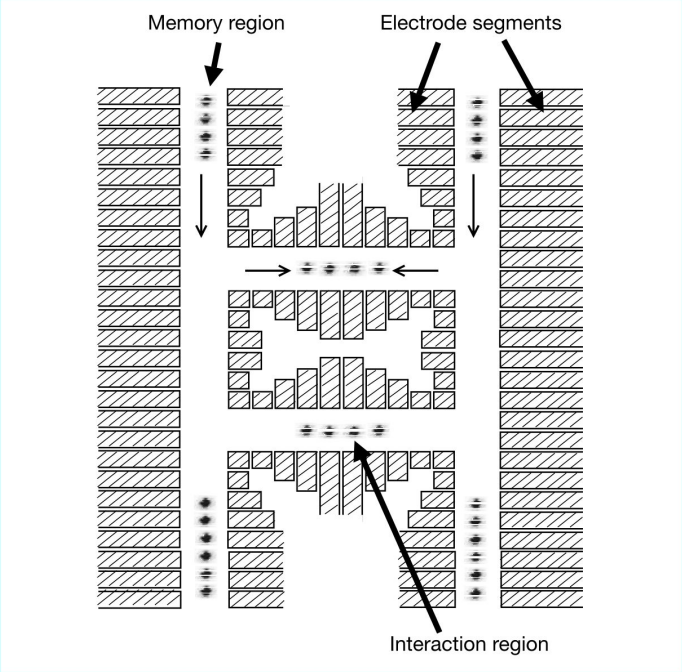
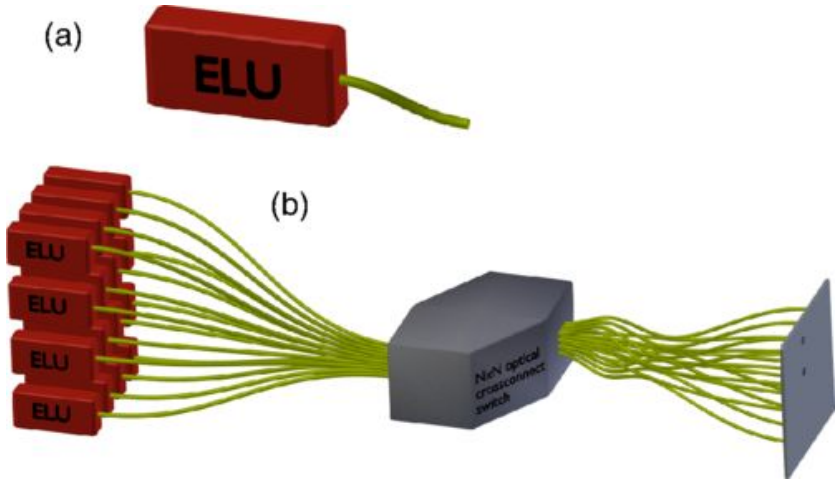


Figure 1 Diagram of the quantum charge-coupled device (QCCD). Ions are stored in the memory region and moved to the interaction region for logic operations. Thin arrows show transport and confinement along the local trap axis.

Implementation of the Gates

Network Model

Multiple small traps are connected by photonic links which enable heralded entanglement.



Heralded Entanglement - entanglement of two ELUs by each emitting a photon that interferes with each other

Elementary Logic Units (Grouping of Qubits) connected together as a network.

Surface-Electrode Traps

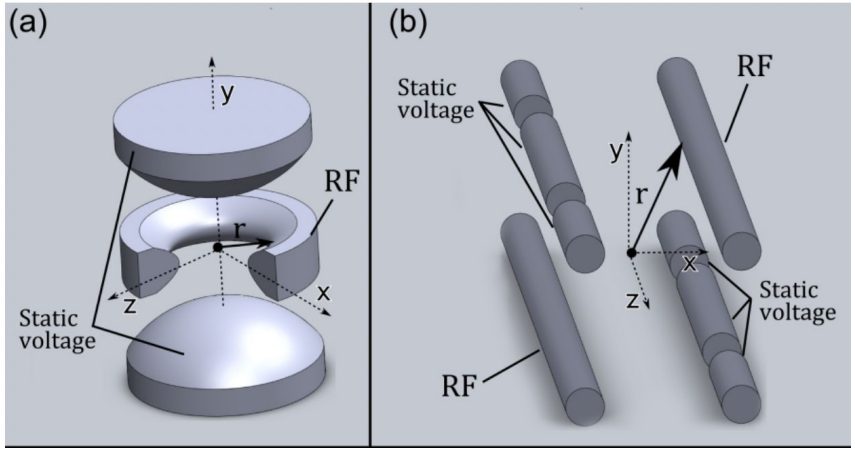


Image: Siverns et al, Applied Phys B (2011)

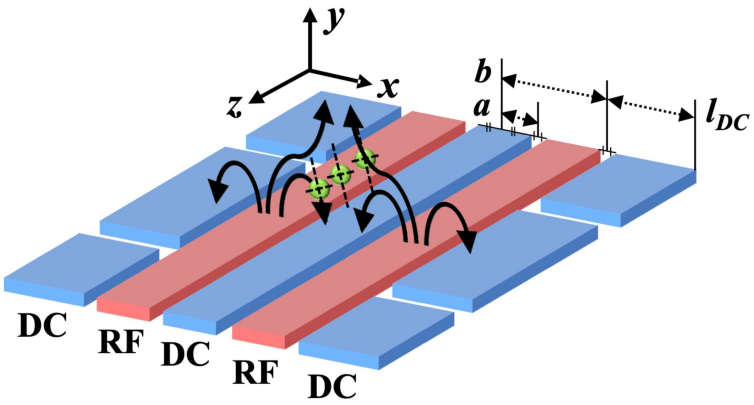


Image: Hong et al, Sensors (2016)

Conclusion

Discussion

- Demonstration of high-fidelity two-qubit gates 99.9%
- Analyzed error sources of the gate. Verified experimental results against theory.
- Characterize the tradeoff between error and gate speed in an experimental setting.

Future Work

- Reduce error further
  - Mitigate magnetic field noise
  - Use better materials
- Utilize integrated photonics to implement the gate
  - Mehta et al. (2020)

Integrated optical multi-ion quantum logic (2020)